HIA-Net: Hierarchical Interactive Alignment Network for Multimodal Few-Shot Emotion Recognition

Yuankang Fu, Kaixiang Yang, *Member, IEEE*, Song Sun, *Member, IEEE*, Xinrong Gong, *Member, IEEE*, and Huanqiang Zeng, *Senior Member, IEEE*

***Abstract*—Physiological multimodal emotion recognition (PMER) has become a key research direction for advancing human-computer interaction and affective computing. However, current PMER methods are affected by significant individual differences and the limited number of samples, making it challenging to capture complex emotional states comprehensively. To address aforementioned issues, this letter proposes a novel multimodal Few-Shot emotion recognition model, called Hierarchical Interactive Alignment Network (HIA-Net). Specifically, the Hierarchical Adaptive Interactive Attention (HAIA) module of HIA-Net is proposed to capture multidimensional emotional features and aggregate the cross-modal information effectively. Additionally, a cross-domain optimization strategy based on the maximum mean discrepancy is proposed to enhance the HIA-Net's adaptability across varying data distributions. Experimental results show that HIA-Net achieves state-of-the-art performance under Few-Shot experimental paradigms on the SEED and SEED-FRA datasets.**

***Index Terms*—EEG, physiological multimodal emotion recognition, few-shot learning, cross-subject, domain adaptation.**

I. Introduction

EMOTIONS are crucial to human cognitive processes and behavior regulation[1], [2], [3]. Neuroscience research shows that emotions involve individual subjective experiences and complex regulation of the central and peripheral nervous systems [4]. Therefore, capturing and analyzing physiological signals can more objectively assess emotional states [5]. However, most current emotion recognition methods rely heavily on single-modal signal sources (e.g., Electroencephalogram (EEG) [6], [7]), which limits the comprehensive analysis and accurate recognition of emotions. Therefore, more attention has been paid to developing effective physiological multimodal emotion recognition methods that integrate various physiological data [8], [9]. For example, Li et al. used a cross-attention mechanism in TACOformer [10], while Chen et al. constructed spatiotemporal graphs in Mul-AT-RGCN [11] to fuse EEG and peripheral signals.

As for central nervous systems, EEG signals are widely used to detect human emotions since their non-invasive, economical, and high sampling rate characteristics [12]. Since the EEG signals can directly reflect the neural activity during emotion induction [13], many researchers have developed EEG-based emotion recognition models, such as hybrid convolutional–transformer architectures [14], multiscale graph convolutional networks [15] and crossdataset pretrained Vision Transformer methods [16]. Moreover, eye movement (EM) signals, as one of the physiological signals from the peripheral nervous system [17], can characterize individuals' gaze behavior and pupil variations under different emotional states [18]. Thus, EM signals can provide supplementary information for the PMER task [19], [20].

Despite the significant progress achieved in PMER, several challenges persist, primarily in the following areas: *(1)Variability in Individual Sample Distributions:* Multimodal physiological signals are non-stationary and exhibit substantial variability among different individuals [21], [22]. This subject-specific variability introduces significant domain shift. Existing domain adaptation methods often overlook the intricate inter-modal interactions and fine-grained hierarchical alignment. *(2) Scarcity of Labeled Samples:* Acquiring labeled emotional data is often costly, and existing PMER models struggle to extract emotional information with limited labeled samples [23], [24].

To tackle the above problems, we present a novel multimodal cross-domain emotion recognition model, named Hierarchical Interactive Alignment Network (HIA-Net), to elegantly reduce the subject differences at multiple granularities and improve the performance under limited labeled sample conditions.

The primary contributions of this study are as follows: 1) The HIA-Net is proposed to enhance adaptability to individual distribution differences in limited emotion-labeled sample scenarios, effectively addressing the challenge of individual variation and data scarcity in the PMER task. 2) We design the Hierarchical Adaptive Interactive Attention Module (HAIA), a novel architecture that jointly performs hierarchical multimodal interaction

Received 8 May 2025; revised 21 June 2025; accepted 25 June 2025. Date of publication 30 June 2025; date of current version 14 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62476101, in part by the Natural Science Foundation of Xiamen under Grant 3502Z202571034, in part by the Fundamental Research Funds for the Central Universities under Grant 2024ZYGXZR062, in part by the Scientific Research Funds of Huaqiao University under Grant 24BS141, and in part by the High-level Talent Team Project of Quanzhou City under Grant 2023CT001. The associate editor coordinating the review of this article and approving it for publication was Dr. John Ball. *(Corresponding author: Xinrong Gong.)*

Yuankang Fu and Kaixiang Yang are with the School of Computer Science and Engineering, South China University of Technology, Guangzhou 510006, China.

Song Sun is with the School of Computer and Information Science, Chongqing Normal University, Chongqing 401331, China.

Xinrong Gong and Huanqiang Zeng are with the School of Engineering, Huaqiao University, Quanzhou 362021, China (e-mail: gongxr@hqu.edu.cn).

Our codes are available at https://github.com/As1yk/HIA-Net.

Digital Object Identifier 10.1109/LSP.2025.3584273

1070-9908 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: National Huaqiao University. Downloaded on January 08,2026 at 03:00:53 UTC from IEEE Xplore. Restrictions apply.

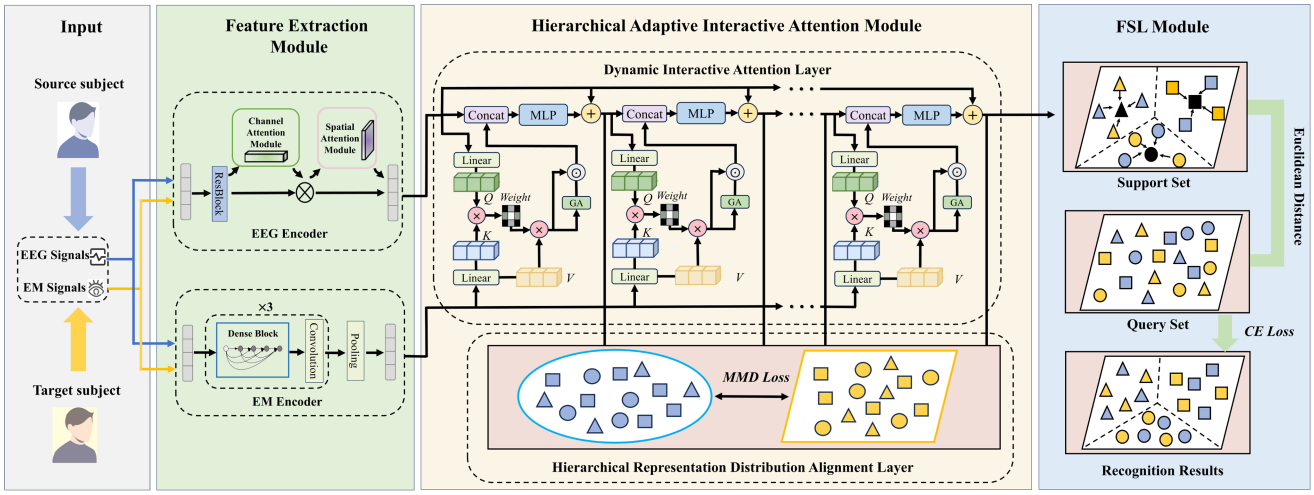


Fig. 1. The general framework of the proposed HIA-Net. It consists of a Feature Extraction Module, a Hierarchical Adaptive Interactive Attention Module for interactive multimodal fusion and domain adaptation, and a FSL Module for Few-Shot emotion classification.

and adaptive domain alignment through a progressive stacking strategy. 3) Extensive experimental results on two benchmark datasets show that HIA-Net obtains state-of-the-art performance in multimodal Few-Shot scenarios.

II. Methodology

A. Problem Definition and Framework

In this work, we cast PMER as a cross-domain adaptation task [25]. The source and target domains refer to data originating from different participants [26]. Let the source domain be $D_s = \{(x_{s1,i}, x_{s2,i}, y_{si})\}_{i=1}^{n_s}$, where each $(x_{s1,i}, x_{s2,i})$ denotes the ith EEG and EM feature pair, and y_{si} is its emotion label. The target domain is split into a small labeled subset $D_{tl} = \{(x_{tl1,i}, x_{tl2,i}, y_{tli})\}_{i=1}^{n_{tl}}$, and a larger unlabeled subset $D_{tu} = \{(x_{tu1,i}, x_{tu2,i})\}_{i=1}^{n_{tu}}$. HIA-Net jointly trains on the source domain D_s and the limited labeled data from the target domain D_{tl}, evaluating its performance on the large unlabeled dataset D_{tu}. The overall framework is illustrated in Fig. 1.

B. Feature Extraction Module

1) EEG Encoder: HIA-Net employs a convolutional neural network combining a residual block [27] with a Convolutional Block Attention Module (CBAM) block [28] for EEG feature extraction. This residual block comprises three convolutional layers with batch normalization, integrating channel and spatial attention mechanisms via the CBAM module. The feature extraction process performed by the encoder can be represented as: $\tilde{H}_G = F^G(X_{EEG})$.

2) EM Encoder: HIA-Net uses DenseNet [29] for EM feature extraction. Its densely connected architecture facilitates feature transfer and reuse, captures hierarchical characteristics in eye movement data, alleviates vanishing gradients. The feature extraction process performed by the encoder can be represented as: $\tilde{H}_M = F^M(X_{EM})$.

C. Hierarchical Adaptive Interactive Attention Module

The Hierarchical Adaptive Interactive Attention Module comprises layers for dynamic interactive attention and hierarchical representation distribution alignment, designed to capture complex cross-modal relationships and effectively align domain distributions across different abstraction levels.

1) Dynamic Interactive Attention Layer (DIA): In the DIA Layer, EEG data forms the query Q_G through linear projection. EM data is similarly mapped to generate the key K_M and value V_M:

$$Q_G = W_Q \tilde{H}_G, \tag{1}$$

$$K_M = W_K \tilde{H}_M, \tag{2}$$

$$V_M = W_V \tilde{H}_M. \tag{3}$$

Scaled dot product attention is applied between EEG queries and EM keys, weighting the EM value matrix to yield fused feature representations:

$$Z = softmax\left(\frac{Q_G K_M^T}{\sqrt{d}}\right) V_M. \tag{4}$$

The Gate Attention (GA) mechanism is proposed to dynamically allocate channel weights that focus on emotionally discriminative features. Channel weights are computed as: $w = \sigma(W_{GA} \cdot Z)$, where σ represents the activation function and W_{GA} denotes the gated weights.

Using channel weights, the GA mechanism performs gated selection on the fused features to obtain the GA-selected features: $Z_{GA} = w \odot Z$, where $\odot$ denotes the element-wise multiplication. Then GA-selected emotional features are concatenated with EEG features and fused through MLP:

$$F_{fused} = MLP(Z_{GA} \oplus \tilde{H}_G), \tag{5}$$

where $\oplus$ represents the concatenation operation. Fused features combine with the original EEG features via a residual connection: $F_{final} = F_{fused} + \tilde{H}_G$.

Authorized licensed use limited to: National Huaqiao University. Downloaded on January 08,2026 at 03:00:53 UTC from IEEE Xplore. Restrictions apply.

The residual connection ensures stable stacking of DIA layers, facilitating capture of incremental multi-level EEG-EM interactions for optimized cross-modal fusion.

2) Hierarchical Representation Distribution Alignment (HRDA): In the HRDA, the maximum mean difference (MMD) loss function is used to adaptively eliminate the inter-domain differences in the multilayer interaction modal information, as defined below:

$$L_{MMD} = \left\| \frac{1}{|X_s|} \sum_{x_s \in X_s} \phi(x_s) - \frac{1}{|X_t|} \sum_{x_t \in X_t} \phi(x_t) \right\|_H^2 . \quad (6)$$

In this context, x_s refers to features derived from the source domain X_s, while x_t signifies features obtained from the target domain X_t. The mapping function $\varphi(\cdot)$ transforms these data points into a Reproducing Kernel Hilbert Space (RKHS), denoted by H.

To dynamically adjust the influence of features at different depths in the MMD loss and prioritize the alignment of deeper features, we propose a hierarchical logarithmic decay weighting mechanism. It is formulated as: $w_i = a \cdot \log(b \cdot i + 1)$, where w_i denotes the weights of the features at level i, and parameters a and b control the logarithmic decay. To ensure that the sum of the weights across layers equals 1, we normalize the weights. The final multilayer MMD loss is: $L_{MMD} = \sum_{i=1}^{N} w_i L_{MMD}^i$, where L_{MMD}^i represents the MMD loss of the i-th layer, and N denotes the total number of layers.

D. Overall Few-Shot Learning Objective

To address the challenges of limited sample size in PMER, this study introduces a Few-Shot learning approach based on class prototypes [30], [31]. In our Few-Shot learning setup, experiments are conducted in episodes, each comprising a support set and a query set. In an N-way K-shot setting, the support set contains $N \times K$ labeled samples, and the query set consists of $N \times Q$ samples for prediction.

1) Prototype Calculation: For each category i, the prototype p_i is computed as the mean fused feature of all support set samples in that category: $p_i = \frac{1}{|S_i|} \sum_{x \in S_i} f(x)$, where S_i represents the samples of category i within the support set, $|S_i|$ is the number of samples in S_i, and $f(x)$ is the fused feature for sample x.

2) Categorization: The probability of each category for a query sample x is computed as:

$$p(y = i|x) = \frac{\exp(-dist(f(x), p_i))}{\sum_{j=1} \exp(-dist(f(x), p_j))}, \quad (7)$$

where $dist(-,-)$ represents the Euclidean distance calculated between the query feature $f(x)$ and the prototype p_i. The classification loss is given: $L_{CE} = -\sum_{i=1}^{N} y_i \log p(y = i|x)$, where N is the number of classes and y_i is the true label of x.

The overall loss function L_{total} for HIA-Net integrates the classification loss L_{CE} and MMD loss L_{MMD} for domain alignment: $L_{total} = L_{CE} + \lambda L_{MMD}$, where λ balances these losses.

TABLE I
SUMMARY OF EMOTIONAL CLASS DISTRIBUTION IN SEED AND SEED-FRA DATASETS

Dataset	Positive	Neutral	Negative	Total
SEED	10008	9864	10440	30312
SEED-FRA	8472	5832	5712	20016

III. EXPERIMENTS

A. Dataset and Processing

HIA-Net is evaluated on two multimodal emotion datasets: SEED and SEED-FRA. These datasets include three emotion categories: *Positive*, *Neutral*, and *Negative*. Emotions are naturally evoked by film clips and ground truth based on participants' subjective reports.

1) SEED [32]: This dataset includes data from 12 Chinese participants collected across three sessions. Each session involved presenting 15 stimulus clips to elicit emotions, resulting in 842 samples per session for each participant.

2) SEED-FRA [33]: This dataset consists of 8 French participants, each undergoing three experimental sessions with 21 stimulus clips per session, yielding 834 samples per session for each participant.

Table I summarizes the distribution of samples across the emotional categories for both datasets.

From the preprocessed EEG signals, differential entropy features were extracted per frequency band and transformed into 9×9 two-dimensional matrices, following the approach in [34]. For EM signals, 33 features were extracted [35], including pupil diameter, blink duration, and other statistical features.

B. Experimental Setup and Compared Methods

Model performance was reported as the average accuracy across the three experimental sessions. For each session, leave-one-subject-out cross-validation was performed independently. Within each session's cross-validation, subjects were alternately designated as the target domain and the remaining subjects as the source domain.

All source domain samples constituted the training set. In the target domain, K labeled samples per class (totaling $N \times K$) were randomly selected for L_{MMD} computation during training, and they constitute the support set for validation and testing. Additionally, 300 samples are used for validation, and the remainder for testing. During training, support and query sets were drawn from the source domain. For validation and testing, the $N \times K$ target domain samples served as the support set, and query sets (each $N \times 20$) were drawn from the target validation or test sets.

In the N-way K-shot setting, N is set to 3 and $K = 1, 5, 10$ or 20. Three DIA layers were used, and the Adam optimizer minimized the loss function with a learning rate of 1×10^{-4}. The loss function's λ was dynamically adjusted as $\lambda = \frac{2}{1+\exp(-\theta p)} - 1$, with $p = \frac{epoch}{nepoch}$ and $\theta = 10$. An early stopping scheme reduced training time over 50 rounds, each with 20 episodes.

The performance of HIA-Net is compared with several representative methods, which can be categorized into two types: 1) Few-Shot learning methods (originally designed for unimodal data but adapted in this study for multimodal physiological inputs), i.e., MatchingNet [36], ProtoNet [30], and MAML [37]; 2) Few-Shot domain adaptation methods, i.e., SDA-FSL [34],

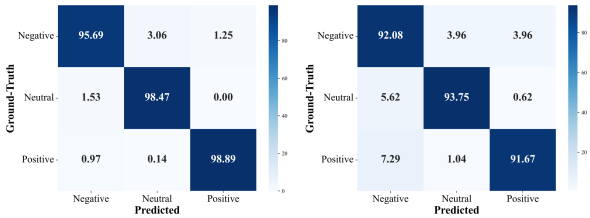
Authorized licensed use limited to: National Huaqiao University. Downloaded on January 08,2026 at 03:00:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II
THE ACCURACY (%) OF THE PMER TASKS ON SEED AND SEED-FRA DATASET

Dataset	Method	1-shot	5-shot	10-shot	20-shot
SEED	MatchingNet(2016)	83.21 ± 1.07	89.18 ± 1.41	90.72 ± 0.49	92.15 ± 0.51
	ProtoNet(2017)	83.93 ± 1.70	91.21 ± 1.43	92.82 ± 1.91	92.84 ± 0.76
	MAML(2017)	84.91 ± 0.42	89.06 ± 1.12	90.92 ± 0.77	93.10 ± 4.56
	MTL-MSRN(2022)	82.07 ± 1.55	82.68 ± 3.00	83.53 ± 2.59	83.74 ± 1.52
	LDP-Net(2023)	81.90 ± 3.35	91.61 ± 1.03	94.00 ± 1.21	95.06 ± 0.85
	SDA-FSL(2021)	82.98 ± 2.10	93.86 ± 1.21	96.13 ± 0.25	96.54 ± 1.16
	HIA-Net	**86.60 ± 2.16**	**95.83 ± 0.85**	**98.34 ± 0.96**	**99.00 ± 0.33**
SEED-FRA	MatchingNet(2016)	66.11 ± 4.14	81.12 ± 1.70	82.09 ± 0.31	84.85 ± 1.98
	ProtoNet(2017)	67.85 ± 3.33	79.84 ± 2.59	85.76 ± 2.51	87.31 ± 2.54
	MAML(2017)	66.67 ± 2.27	80.25 ± 0.57	83.22 ± 3.01	88.51 ± 0.78
	MTL-MSRN(2022)	68.71 ± 3.61	69.84 ± 0.74	70.98 ± 3.83	71.62 ± 1.15
	LDP-Net(2023)	64.83 ± 2.03	80.42 ± 3.12	83.47 ± 2.42	86.55 ± 3.38
	SDA-FSL(2021)	66.79 ± 0.43	82.46 ± 1.19	86.04 ± 1.22	89.80 ± 2.19
	HIA-Net	**71.64 ± 1.65**	**83.69 ± 1.79**	**91.17 ± 1.64**	**95.19 ± 2.29**

TABLE III
ABLATION EXPERIMENTS OF THE HIA-NET ON THE SEED AND SEED-FRA DATASETS UNDER THE 1-SHOT SETTING

Method	SEED	SEED-FRA
baseline	81.65 ± 2.24	64.86 ± 1.67
HIA-Net (w/o DIA)	84.13 ± 2.56	66.58 ± 3.15
HIA-Net (w/o HRDA)	82.89 ± 2.32	67.75 ± 2.12
HIA-Net (Ordinary MMD)	84.21 ± 0.56	69.39 ± 2.79
HIA-Net	**86.60 ± 2.16**	**71.64 ± 1.65**



(a) SEED (b) SEED-FRA

Fig. 2. Confusion matrices of HIA-Net on the SEED and SEED-FRA datasets. (a) SEED. (b) SEED-FRA.

MTL-MSRN [38], and LDP-Net [39]; These methods were selected based on their relevance to Few-Shot learning and domain adaptation tasks, representing key approaches and recent advancements in these fields.

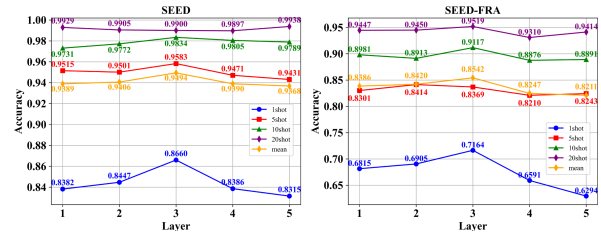
C. Physiological Multimodal Emotion Recognition Results

Table II reports the results. Across various datasets and K-shot settings, HIA-Net obtains the best performance, indicating that it effectively integrates information from both EEG and EM modalities, significantly reducing individual differences among subjects under the limited labeled sample conditions.

D. Analysis and Discussion

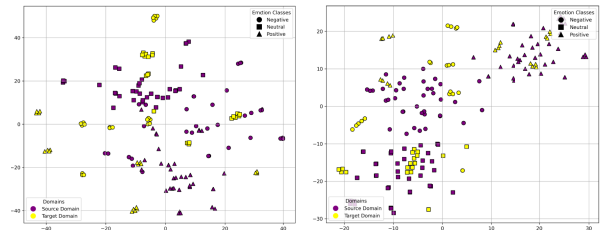
1) Module-Wise Ablation Study: To evaluate the critical roles of the DIA layer and the HRDA in HIA-Net, we conduct an ablation study under the 1-shot setting on the SEED and SEED-FRA datasets. The baseline represents HIA-Net with both the DIA layer and the HRDA layer removed. The results in Table III demonstrate that both the DIA layer and the HRDA layer significantly enhance the model's performance. An ablation study where our HRDA layer is replaced by the application of ordinary MMD loss further highlights HRDA's unique contribution and superior efficacy.

2) Confusion Matrix: Fig. 2 presents confusion matrices for the 10-shot setting. In the SEED dataset, HIA-Net exhibits strong



(a) SEED (b) SEED-FRA

Fig. 3. Comparison of HIA-Net with different DIA layers. (a) SEED. (b) SEED-FRA.



(a) without HRDA (b) with HRDA

Fig. 4. t-SNE visualizations of the fused modal representations learned by HIA-Net: (a) without HRDA, and (b) with HRDA.

classification performance across all three emotion categories. HIA-Net also performs well on the SEED-FRA dataset. It slightly shows confusion between negative and positive emotions, likely due to the smaller sample size of the SEED-FRA.

3) Discussion on the Impact of Multiple Dynamic Interactive Attention Layers: This section assesses the impact of different numbers of DIA layers on two datasets. As illustrated in Fig. 3, increasing the number of DIA layers initially boosts model performance, suggesting that additional layers enhance cross-modal information capture. However, performance declines when the number of layers reaches four. This decline is likely due to increased model complexity and overfitting. This result underscores the effectiveness and flexibility of the HIA-Net design, which is not solely reliant on the number of DIA layers.

4) Visualization of the Embedding Space: Fig. 4 presents t-SNE visualizations [40] of the fused modal representations, which are the final DIA layer output, comparing models trained with and without the HRDA. In contrast, the right panel demonstrates significantly minimized domain differences, showcasing HRDA's effectiveness in tackling distribution mismatch and promoting cross-domain transfer.

IV. CONCLUSION

In this letter, we carefully propose the HIA-Net, a novel framework for the multimodal Few-Shot emotion recognition. HIA-Net employs a multilayer dynamic interaction mechanism to thoroughly explore their complementary features, while the HRDA effectively minimizes modality discrepancies. Furthermore, HIA-Net exhibits exceptional performance in its prototype-based Few-Shot learning approach. In future work, we aim to enhance the prototype representation [41] and privacy protection [42] within HIA-Net to achieve improved performance.

Authorized licensed use limited to: National Huaqiao University. Downloaded on January 08,2026 at 03:00:53 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] C. E. Izard, "Forms and functions of emotions: Matters of emotion–Cognition interactions," *Emotion Rev.*, vol. 3, no. 4, pp. 371–378, Sep. 2011.

[2] V. De Pascalis and C. Moret-Tatay, "Editorial: Emotion regulation and cognitive processes," *Front. Hum. Neurosci.*, vol. 17, 2023, Art. no. 1250666.

[3] P. Valenzuela, C. Navarini, J. A. Mercado, B. Fowers, and A. Panno, "Editorial: Emotional regulation and human flourishing: Theoretical and empirical perspectives," *Front. Psychol.*, vol. 16, 2025, Art. no. 1565130.

[4] R. W. Levenson, "The autonomic nervous system and emotion," *Emotion Rev.*, vol. 6, no. 2, pp. 100–112, Mar. 2014.

[5] K. R. Scherer and T. Bänziger, "On the use of actor portrayals in research on emotional expression," in *Blueprint for Affective Computing: A Sourcebook and Manual*. Oxford, U.K.: Oxford Univ. Press, 2010, pp. 166–176.

[6] X. Zhang and X. Cheng, "A transformer convolutional network with the method of image segmentation for EEG-based emotion recognition," *IEEE Signal Process. Lett.*, vol. 31, pp. 401–405, 2024.

[7] V. Lawhern, A. Solon, N. Waytowich, S. Gordon, C. Hung, and B. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain–Computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, 2018, Art. no. 056013.

[8] N. Ahmed, Z. Al Aghbari, and S. Girija, "A systematic survey on multimodal emotion recognition using learning algorithms," *Intell. Syst. Appl.*, vol. 17, Feb. 2023, Art. no. 200171.

[9] S. Kalateh, L. A. Estrada-Jimenez, S. Nikghadam-Hojjati, and J. Barata, "A systematic review on multimodal emotion recognition: Building blocks, current state, applications, and challenges," *IEEE Access*, vol. 12, pp. 103976–104019, 2024.

[10] X. Li, "TACOformer: Token-channel compounded cross attention for multimodal emotion recognition," 2023, *arXiv:2306.13592*.

[11] J. Chen, Y. Liu, W. Xue, K. Hu, and W. Lin, "Multimodal EEG emotion recognition based on the attention recurrent graph convolutional network," *Information*, vol. 13, no. 11, 2022, Art. no. 550.

[12] E. Gkintoni, A. Aroutzidis, H. Antonopoulou, and C. Halkiopoulos, "From neural networks to emotional networks: A systematic review of EEG-based emotion recognition in cognitive neuroscience and real-world applications," *Brain Sci.*, vol. 15, no. 3, 2025, Art. no. 220.

[13] W. Ma, Y. Zheng, T. Li, Z. Li, Y. Li, and L. Wang, "A comprehensive review of deep learning in EEG-based emotion recognition: Classifications, trends, and practical implications," *PeerJ Comput. Sci.*, vol. 10, 2024, Art. no. e2065.

[14] R. Liu et al., "ERTNet: An interpretable transformer-based framework for EEG emotion recognition," *Front. Neurosci.*, vol. 18, 2024, Art. no. 1320645.

[15] L. Cao, W. Zhao, and B. Sun, "Emotion recognition using multi-scale EEG features through graph convolutional attention network," *Neural Networks*, vol. 184, 2025, Art. no. 107060.

[16] F. Wang, Y.-C. Tian, and X. Zhou, "Cross-dataset EEG emotion recognition based on pre-trained vision transformer considering emotional sensitivity diversity," *Expert Syst. Appl.*, vol. 279, Jun. 2025, Art. no. 127348.

[17] M. M. Bradley, L. Miccoli, M. A. Escrig, and P. J. Lang, "The pupil as a measure of emotional arousal and autonomic activation," *Psychophysiol.*, vol. 45, no. 4, pp. 602–607, Jul. 2008.

[18] X. Gong, C. L. P. Chen, and T. Zhang, "Cross-cultural emotion recognition with EEG and eye movement signals based on multiple stacked broad learning system," *IEEE Trans. Comput. Social Syst.*, vol. 11, no. 2, pp. 2014–2025, Apr. 2024.

[19] C. Fei, R. Li, L. Zhao, Z. Li, and B.-L. Lu, "A cross-modality deep learning method for measuring decision confidence from eye movement signals," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Glasgow, U.K., 2022, pp. 3342–3345.

[20] B. Fu, C. Gu, M. Fu, Y. Xia, and Y. Liu, "A novel feature fusion network for multimodal emotion recognition from EEG and eye movement signals," *Front. Neurosci.*, vol. 17, 2023, Art. no. 1234162.

[21] Y. Zhou et al., "Cross-subject cognitive workload recognition based on EEG and deep domain adaptation," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 2518912.

[22] X. Gong, C. L. P. Chen, B. Hu, and T. Zhang, "CiABL: Completeness-induced adaptative broad learning for cross-subject emotion recognition with EEG and eye movement signals," *IEEE Trans. Affect. Comput.*, vol. 15, no. 4, pp. 1970–1984, Oct.–Dec. 2024.

[23] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, "A comprehensive survey of few-shot learning: Evolution, applications, challenges, and opportunities," *ACM Comput. Surv.*, vol. 55, no. 13s, 2023, Art. no. 271.

[24] F. Qi, H. Zhang, X. Yang, and C. Xu, "A versatile multimodal learning framework for zero-shot emotion recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5728–5741, Jul. 2024.

[25] M. Jimenez-Guarneros and G. Fuentes-Pineda, "CFDA-CSF: A multimodal domain adaptation method for cross-subject emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 15, no. 3, pp. 1502–1513, Jul.–Sep. 2024.

[26] X. Wu, X. Ju, S. Dai, X. Li, and M. Li, "Multi-source domain adaptation for EEG emotion recognition based on inter-domain sample hybridization," *Front. Hum. Neurosci.*, vol. 18, 2024, Art. no. 1464431.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.

[28] S. Woo, J. Park, Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., 2018, pp. 3–19.

[29] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2261–2269.

[30] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, vol. 30, pp. 4080–4090.

[31] Y. Liu, D. Shi, and H. Lin, "Few-shot learning with representative global prototype," *Neural Networks*, vol. 180, Dec. 2024, Art. no. 106600.

[32] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Ment. Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.

[33] W. Liu, W. L. Zheng, Z. Li, S. Y. Wu, L. Gan, and B. L. Lu, "Identifying similarities and differences in emotion recognition with EEG and eye movements among Chinese, German, and French people," *J. Neural Eng.*, vol. 19, no. 2, Mar. 2022, Art. no. 026012.

[34] R. Ning, C. L. Philip Chen, and T. Zhang, "Cross-subject eeg emotion recognition using domain adaptive few-shot learning networks," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, Houston, TX, USA, 2021, pp. 1468–1472.

[35] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki, "EmotionMeter: A multimodal framework for recognizing human emotions," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 1110–1122, Mar. 2019.

[36] O. Vinyals, C. Blundell, T. Lillicrap, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, vol. 29, pp. 3637–3645.

[37] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, 2017, vol. 70, pp. 1126–1135.

[38] J. Li et al., "Cross-subject EEG emotion recognition combined with connectivity features and meta-transfer learning," *Comput. Biol. Med.*, vol. 145, Jun. 2022, Art. no. 105519.

[39] F. Zhou, P. Wang, L. Zhang, W. Wei, and Y. Zhang, "Revisiting prototypical network for cross domain few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Vancouver, BC, Canada, 2023, pp. 20061–20070.

[40] L. van der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, Jan. 2014.

[41] W. Fu, L. Zhou, and J. Chen, "Bidirectional matching prototypical network for few-shot image classification," *IEEE Signal Process. Lett.*, vol. 29, pp. 982–986, 2022.

[42] X. Gong et al., "Adaptive compressed-based privacy-preserving large language model for sensitive healthcare," *IEEE J. Biomed. Health Inform.*, early access, Apr. 08, 2025, doi: 10.1109/JBHI.2025.3558935.

Authorized licensed use limited to: National Huaqiao University. Downloaded on January 08,2026 at 03:00:53 UTC from IEEE Xplore. Restrictions apply.